PART 1.

Assessment of Stevens Flood Advisory System (SFAS) service and predictions for Nor'Easter Stella

Four days before the event, between 9am and 12pm EST March 10th 2017, automated Flood Watch notices were sent to SFAS subscribers, warning them of a coming flood event. These notices were issued for a total of 45 stations (Table 1). Most notices were followed next day with a 3-day out flood watch notice.

Table 1: SFAS 4-day Flood Watches issued for Nor'Easter Stella on March 10, 2017. Watches for which a 3-day Flood Advisory was also issued are highlighted in Bold to show consistency of early forecasts. Short-term Flood Advisories issued in advance of peak heights the day of the storm, March 14, 2017, are underlined. Automated Early Flood Watches were issued for a total of 45 coastal stations, with a 30-station subset included in Automated Flood Advisory e-mails the day of the storm. URL color shows the verifying NWS local flood conditions: Above Minor (Yellow) or Above Moderate Flood Level (Orange).

FDU Bridge at Hackensack NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=MEEMS_1

Saw Mill Creek at Kearny NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=MEEMS_2

Kearny Point at Kearny NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=MEEMS_3

Barge Marina at Carlstadt NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=MEEMS_4

New Haven CT: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N010

Bridgeport CT: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N011

Kings Point NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N016

Bergen Point West Reach NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N019

Sandy Hook NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N021

Atlantic City NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N022

Cape May NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N023

Lewes DE: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N030

Ocean City Inlet MD: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N031

Windmill Point VA: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N051

Chesapeake Bay Bridge Tunnel VA: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N058

Duck NC: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N061

Elizabeth Channel near Newark NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=SPEWRAP2

Arthur Kill near Bloomfield NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=SPHHPIP1

Hudson River near Jersey City NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=SPNJTLP1

Bowery Bay at East Elmhurst NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=SPLGAAP1

Passaic River near Harrison NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=SPPATHP2

Newark Bay near Elizabeth NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=SPPNPEP1

Great South Bay at Lindenhurst NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U204

Hudson Bay at Freeport NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U205

Reynolds Channel at Point Lookout NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U206

East Rockaway Inlet at Atlantic Beach NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U207

Rockaway Inlet near Floyd Bennett Field NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U209

Hackensack River at Hackensack NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U215

Passaic River at PVSC at Newark NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U216

Raritan River at South Amboy NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U217

Raritan Bay at Keansburg NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U218

Shrewsbury River at Sea Bright NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U219

Shark River at Belmar NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U220

Barnegat Bay at Barnegat Light NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U225

Little Egg Inlet near Tuckerton NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U227

Absecon Creek Rte 30 at Absecon NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U228

Inside Thorofare Rte 40 Atlantic City NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U229

Ludlam Thorofare at Sea Isle City NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U233

Peck Bay at Ocean City NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U231

Great Channel at Stone Harbor NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U235

Maurice River at Bivalve NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U238

Cohancey River at Greenwich NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U239

Watson Creek at Manasquan NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U249

At 6pm, the same day, these flood watches were issued:

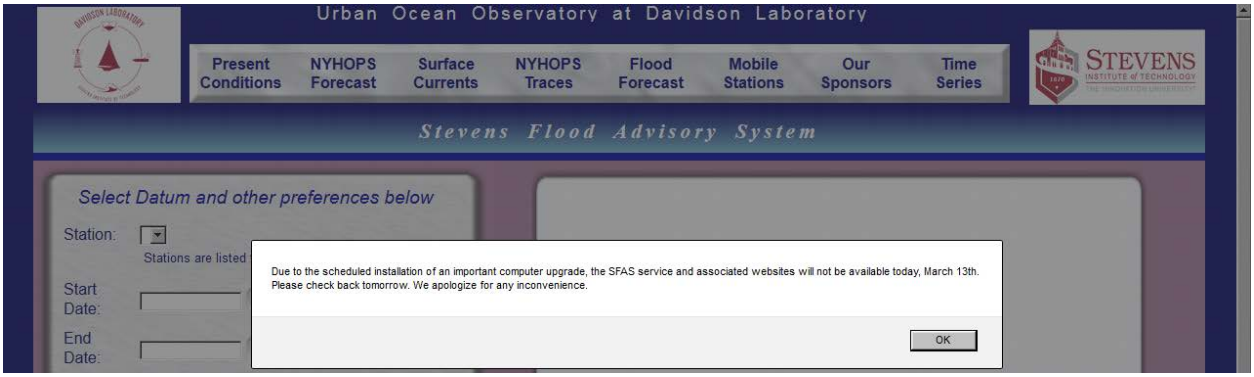
Norrie Point NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=HRNOPTH

Tivoli Bays South NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=HRTVBSH

The automatically -mailed Flood Watch message from SFAS read:

There is a possibility of significant flooding (water levels rising above moderate flood stage) for the station(s) listed below 4 days out from now. We suggest that you use the link for each station below to check the predictions, updated every 6 hours.

After the switch from EST to EDT on Sunday, March 12th, SFAS had to be taken down for a thrice-rescheduled InfiniBand networking upgrade at 8:30am EDT on March 13th. At that point we were in contact with NWS Upton (OKX), sharing notes offline. We were also in contact with other agencies like the City of Hoboken, Monmouth County, the Port Authority of New York and New Jersey, and New Jersey Transit, for specific local information, including providing them with 3m horizontal resolution maps of flooding potential for critical facilities, as per our ongoing flood resiliency projects.



Limited public SFAS service was restored March 14th at 3am EDT. Full SFAS service was restored at 8am EDT that day, the day of the storm.

SFAS Flood Advisory Notices were automatically sent to subscribers at 6am and 12pm EDT, March 14th. Stations for which these short-term Flood Advisories were sent are underlined on Table 1 at the top of the document. Flood Advisories were sent for a total of 30 stations. Importantly, and as seen on Table 1, Flood Watches had also been issued for all these 30 stations four days in advance of the storm, so ample warning existed. 15 other stations received early Flood Watches, but no Flood Advisories the day of the storm (Table 1). Thus, a total of 45 out of 76 SFAS coastal stations received a Flood Watch or Advisory. For almost all stations, the 5%-to-95% ensemble uncertainty shown in SFAS four days prior to the event enveloped the event's local observed peak water level. The 4-day-forecast cross-station average probability of exceedance of the verified March 14th event peaks was 20% (the event verified at the 80% percentile of the 96hour early prediction envelope).

An example of the e-mailed SFAS Flood Advisories and the information they contained is given below:

We expect flooding for the station(s) listed below at some point in the eight-hour period between 2017-03-14 6:00 AM and 2017-03-14 2:00 PM local time. We have indicated the approximate time when we expect the first event to occur for each station. Note that there may be higher flooding later in the eight-hour period, and we suggest that you click on the link for each station below to see complete flooding information for that station.

FDU Bridge at Hackensack NJ at approximately: 2017-03-14 12:15 PM
http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=MEEMS_1

Of the 30 issued Flood Advisories, 16 verified as minor floods, 13 verified as moderate floods, while Ocean City Inlet MD turned out to be a false alarm. Three more stations listed on Table 1 were misses: Minor flooding was observed at Arthur Kill near Bloomfield NY, Ludlam Thorofare at Sea Isle City NJ, and Maurice River at Bivalve NJ, and while an early Flood Watch was issued for these stations (Table 1), no short-term Flood Advisory was issued the day of the storm. Three misses for which no Flood Watch or Advisory were issued (not listed on Table 1) occurred at The Battery NY, at Jamaica Bay near Inwood NY, and at Hudson River near Hoboken NJ; Minor Flood Levels were exceeded by 0.4ft, 0.3ft, and 0.2ft, respectively. Three more near-misses were at New York Harbor near Red Hook NY, New London CT, and Montauk NY where levels just exceeded Minor Flood, by less than 0.1ft. A summary is found on Table 2. These 6 misses were less than 0.4ft above Minor Flood levels and did not experienced above-Moderate-level flooding, which means that the early predictions were partially verified: The uncertainty envelope of the 4- and 3-days-prior predictions did not include a possibility of above moderate flooding at any of these stations, which is the condition required for SFAS to automatically issue a Flood Watch; thus, appropriately, there was no Flood Watch issued for these 6 stations.

Table 2. Summary of SFAS Flood Advisories for Nor'Easter Stella

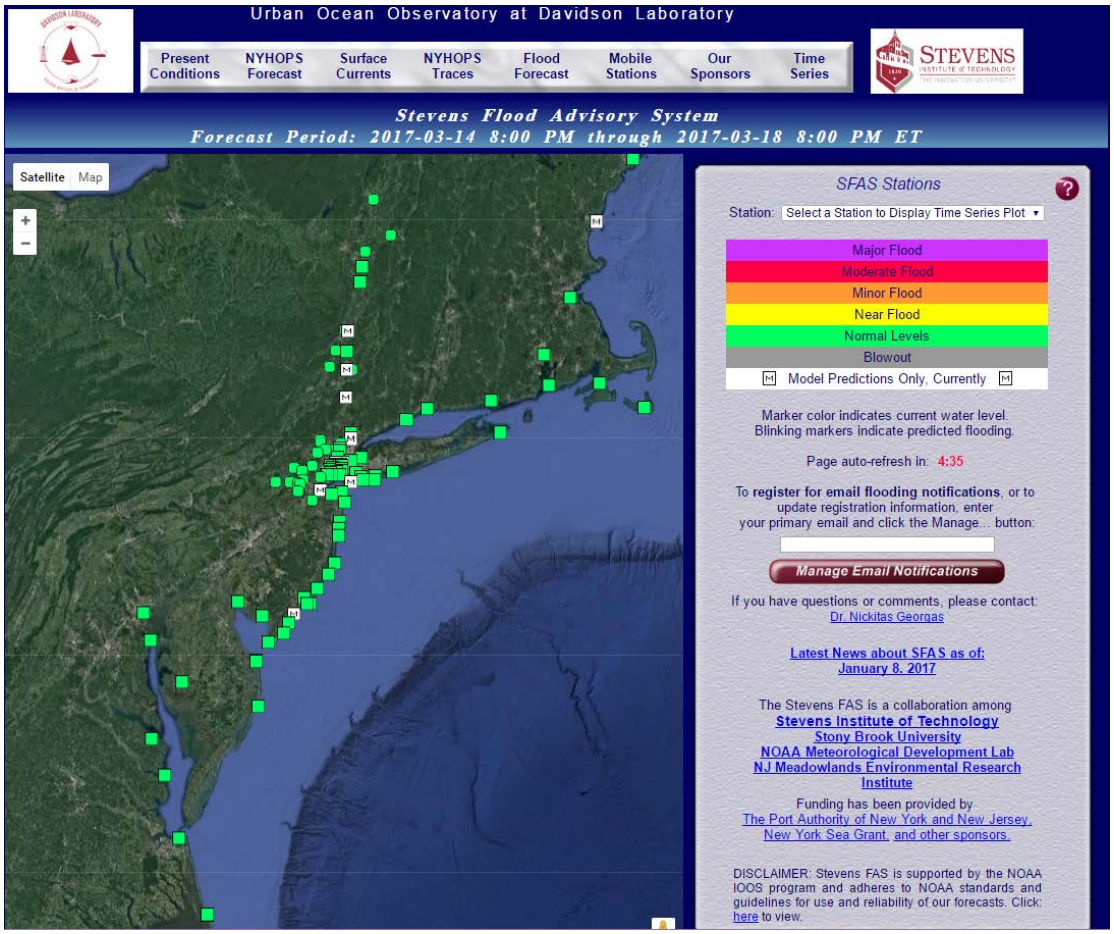
Hits: 29 (stations that received both early flood watches and day-of-the-storm advisories, and flooded) **False Alarms:** 1 (Ocean City Inlet, MD; received advisory but no flood occurred). **Misses:** 9 (stations flooded, but no advisory was issued; 3 received an early Flood Watch, while 6 were not warned against Minor Flooding of 0.1-0.4ft).

PART 2.

Snapshots of storm surges experienced during

Nor'Easter Stella

March 14 2017



Sample Stations:

- The Battery NY
- Kings Point NY
- Sandy Hook NJ
- Atlantic City NJ
- New Haven CT
- Boston MA
- Newport RI
- Lewes DE

How to read the graphs of the Stevens Flood Advisory System

In the figure for each station:

- **The graph on top left corner** includes basic information for each station, such as station's name, start date and end date, datum, forecast model displayed, unit and time zone.
- **The graph on bottom left corner** is the map of the location of the selected station.
- **The graph on top right corner shows water level information:**

  ~Predicted water level based on Stevens Ensemble Forecast models (line)

  ~Ensemble uncertainty from 5% to 95% (Gray area)

  ~Observed water level (Red dots, where available)

  ~Time marker when the snapshot was taken (vertical black dash line)

  Note: For Boston MA station, the Predicted water level is based on Stevens SNAP Extended Ensemble Forecast model (dark green line); the Stevens NYHOPS Ensemble is used for the rest of the stations (pink line)

- **The graph on bottom right corner shows storm surge information:**

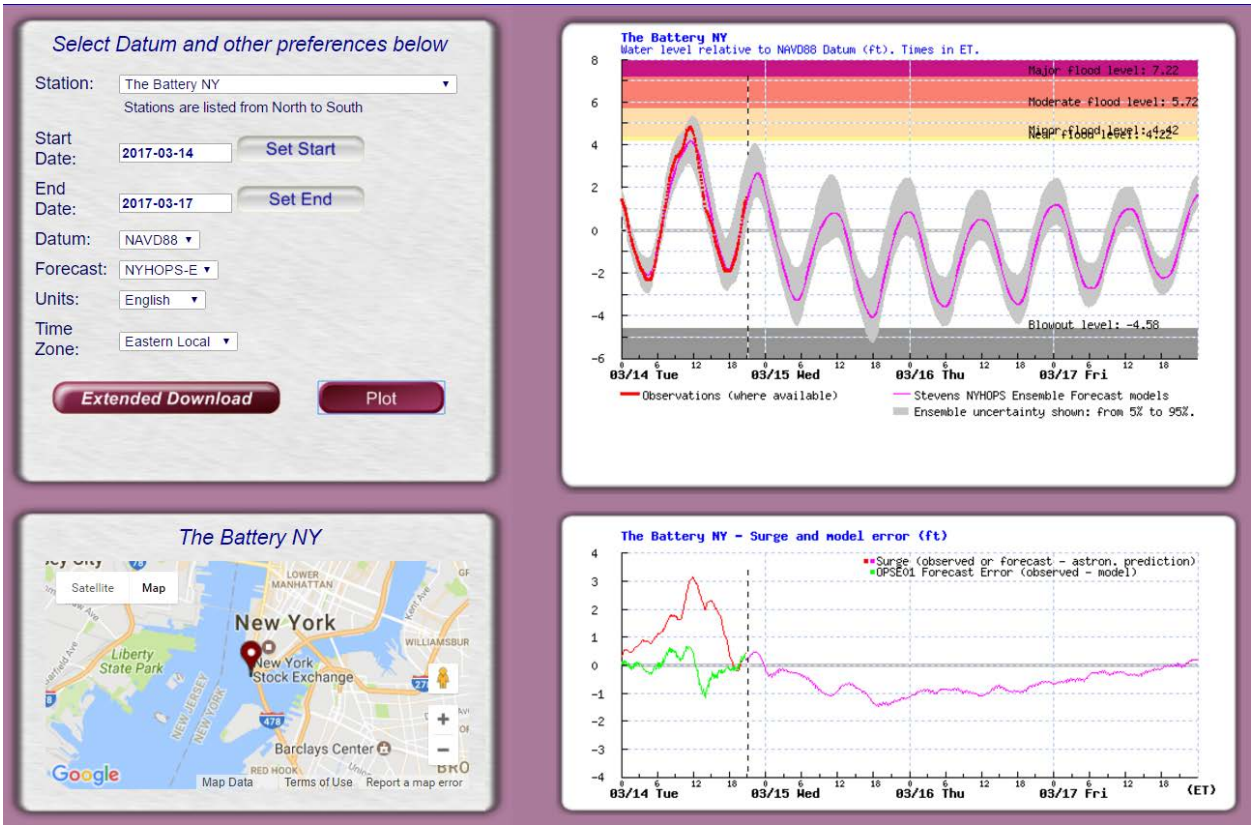
  ~Observed surge = observed water level – predicted astronomical tide (Red line)

  ~Forecast surge = predicted water level – predicted astronomical tide (Pink line)

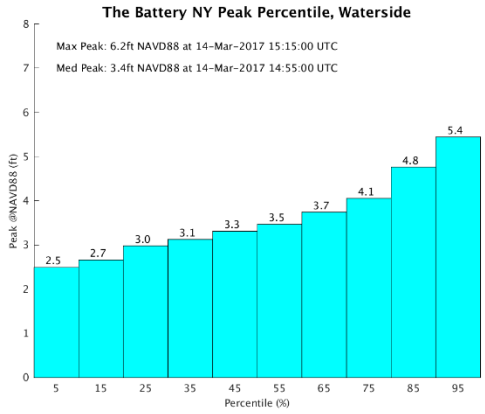
  ~Forecast error = observed surge – model surge (Green line)

  ~Time marker when the snapshot was taken (vertical black dash line)

Figures for sample stations:

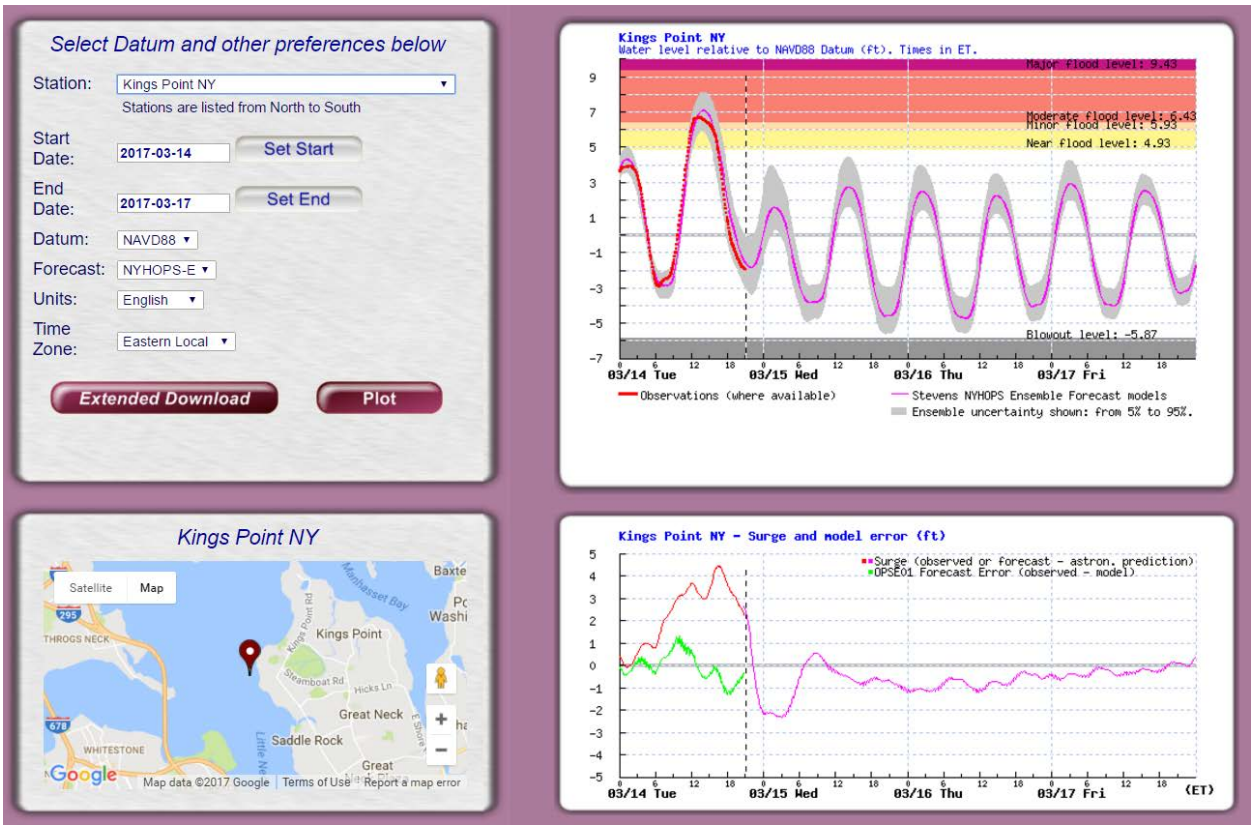


The Battery NY

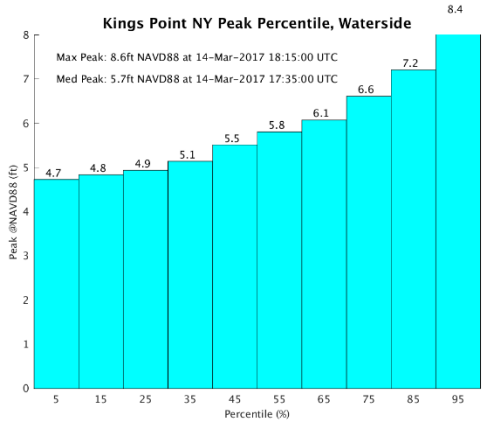


Percentiles of 96hr-prior forecast peaks.

Verified peak was 4.84ft NAVD88 on March 14 2017, 15:18 UTC.

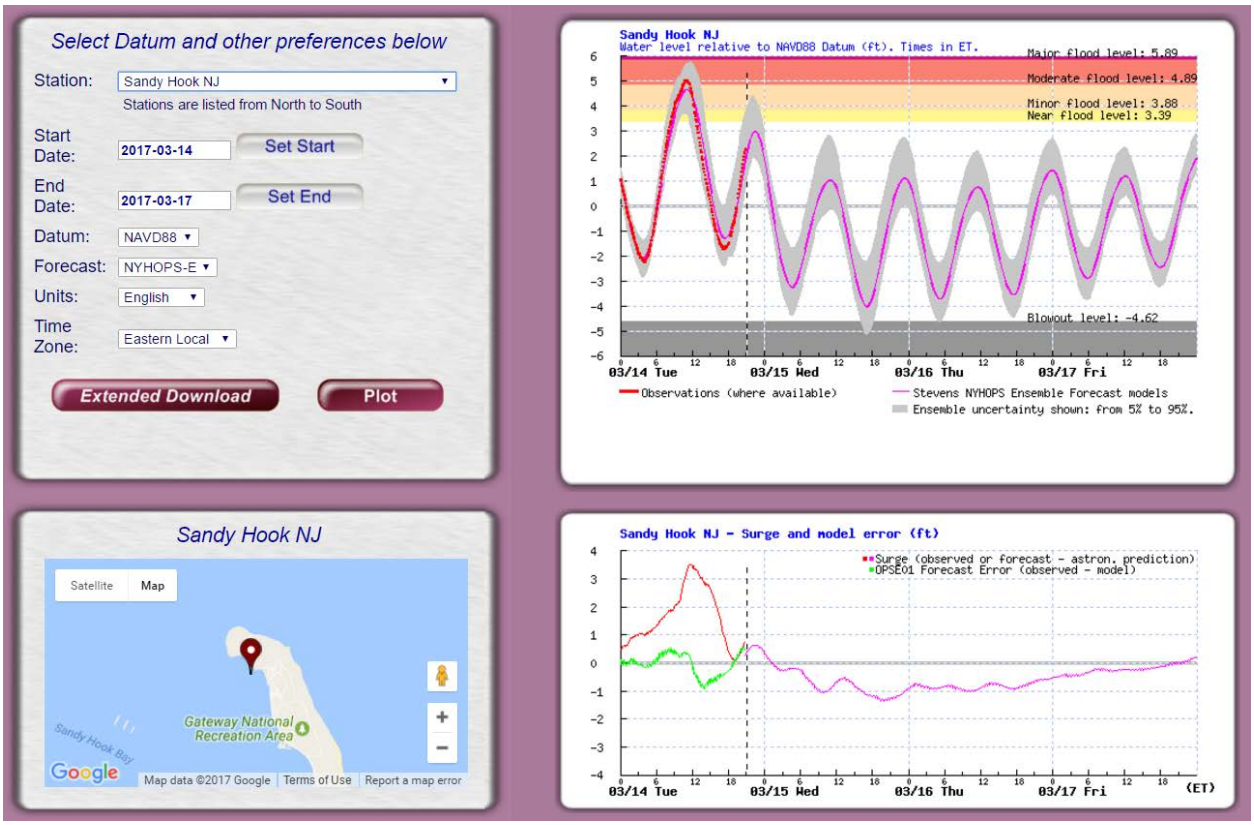


Kings Point NY

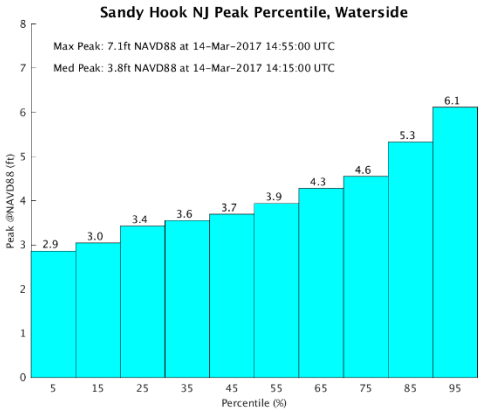


Percentiles of 96hr-prior forecast peaks.

Verified peak was 6.76ft NAVD88 on March 14 2017, 17:12 UTC.

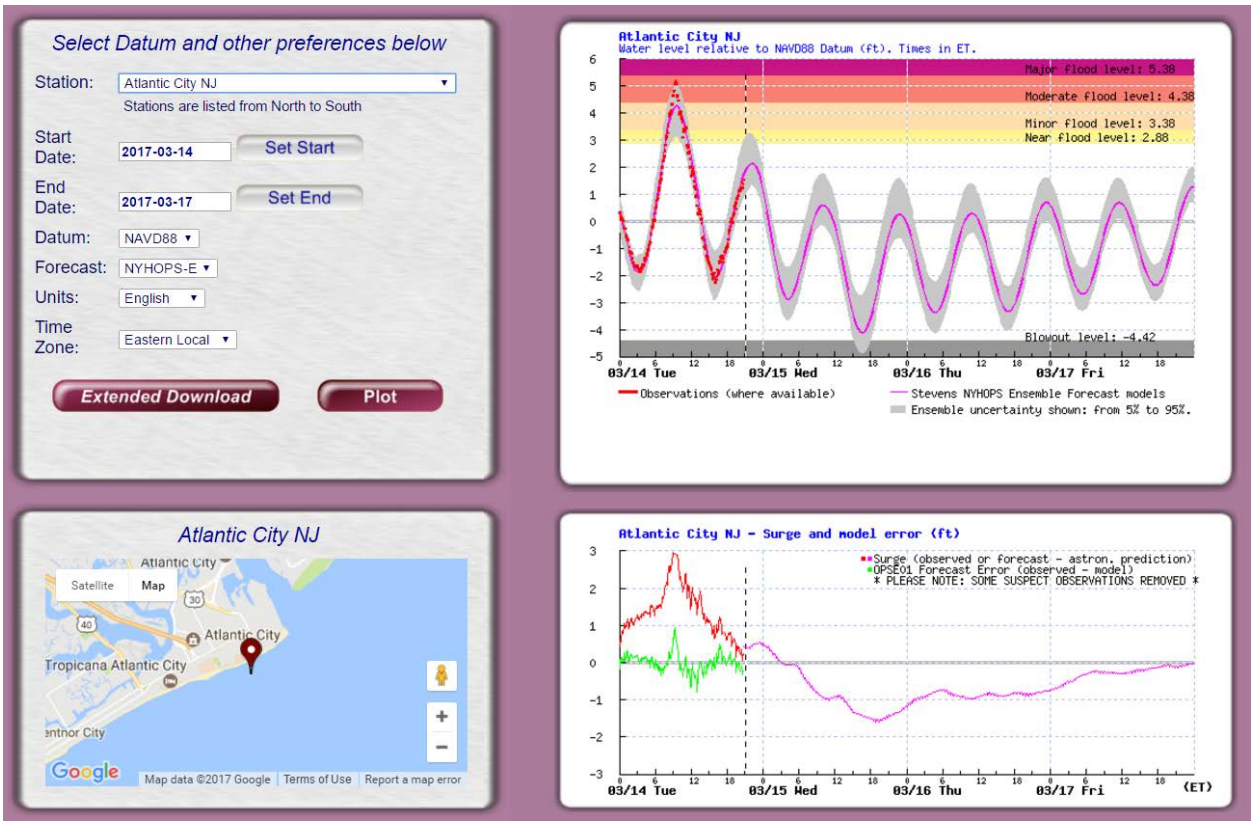


Sandy Hook NJ

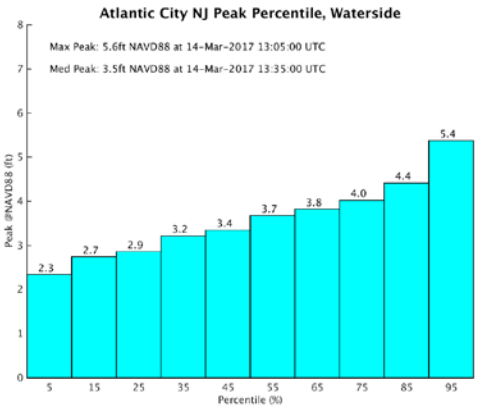


Percentiles of 96hr-prior forecast peaks.

Verified peak was 5.06ft NAVD88 on March 14 2017, 14:42 UTC.

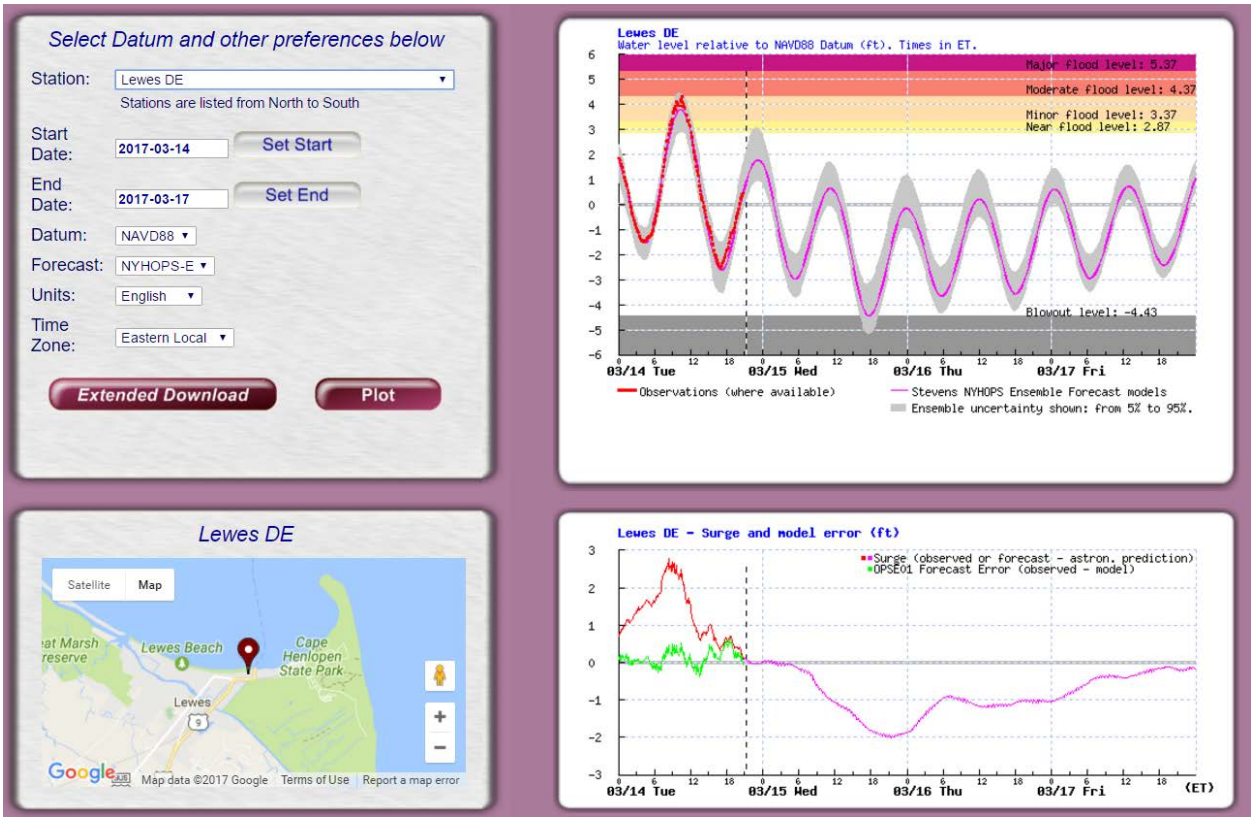


Atlantic City NJ

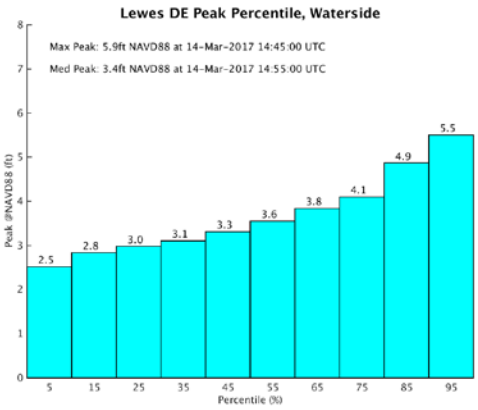


Percentiles of 96hr-prior forecast peaks.

Verified peak was 5.18ft NAVD88 on March 14 2017, 13:18 UTC.

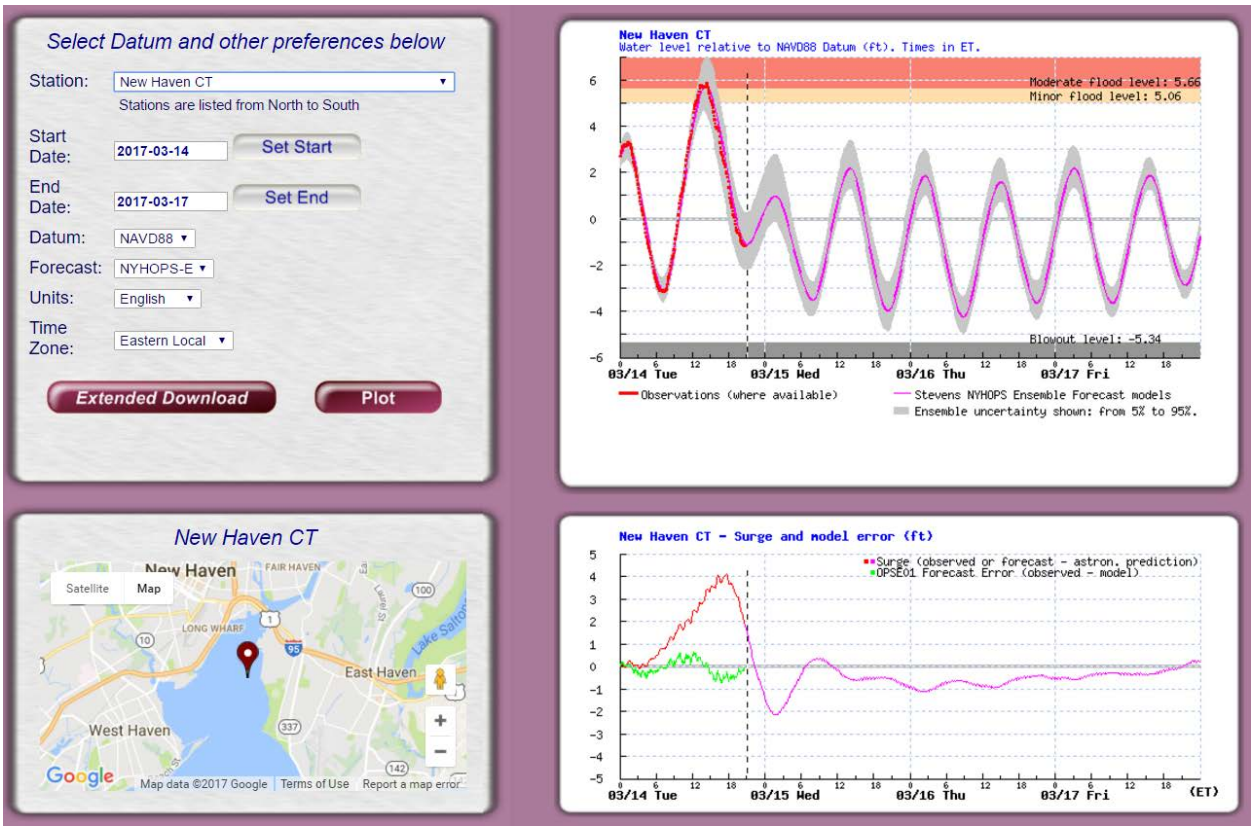


Lewes DE

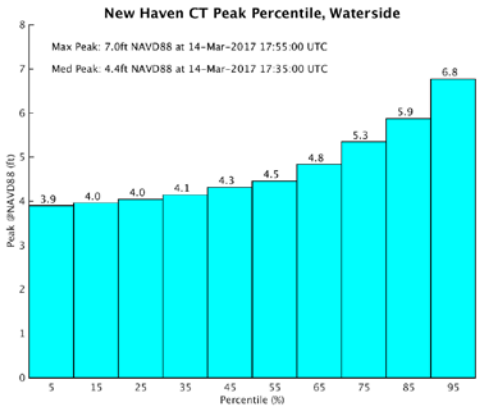


Percentiles of 96hr-prior forecast peaks.

Verified peak was 4.35ft NAVD88 on March 14 2017, 14:30 UTC.

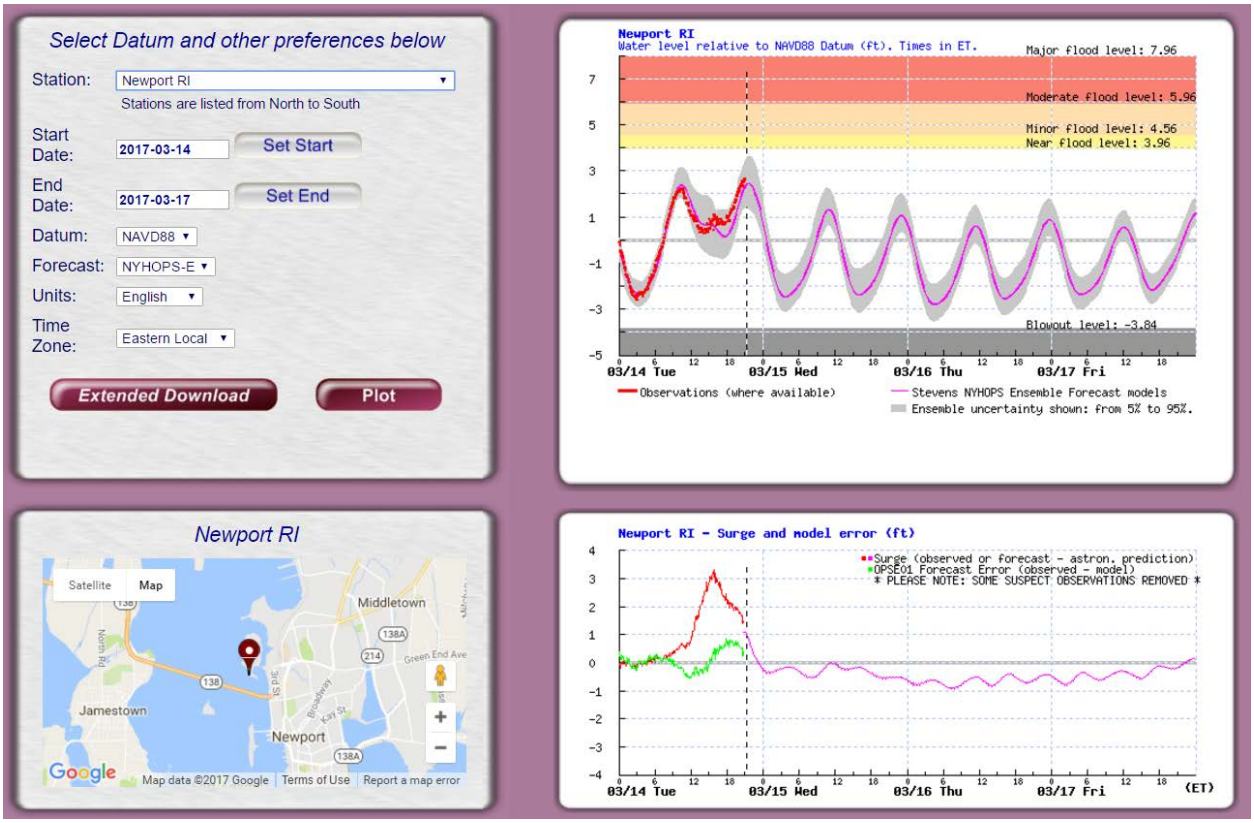


New Haven CT

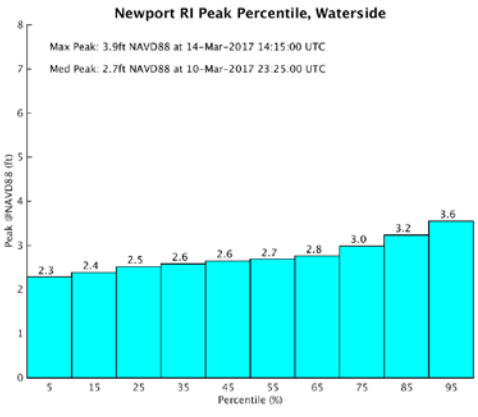


Percentiles of 96hr-prior forecast peaks.

Verified peak was 5.86ft NAVD88 on March 14 2017, 18:18 UTC.

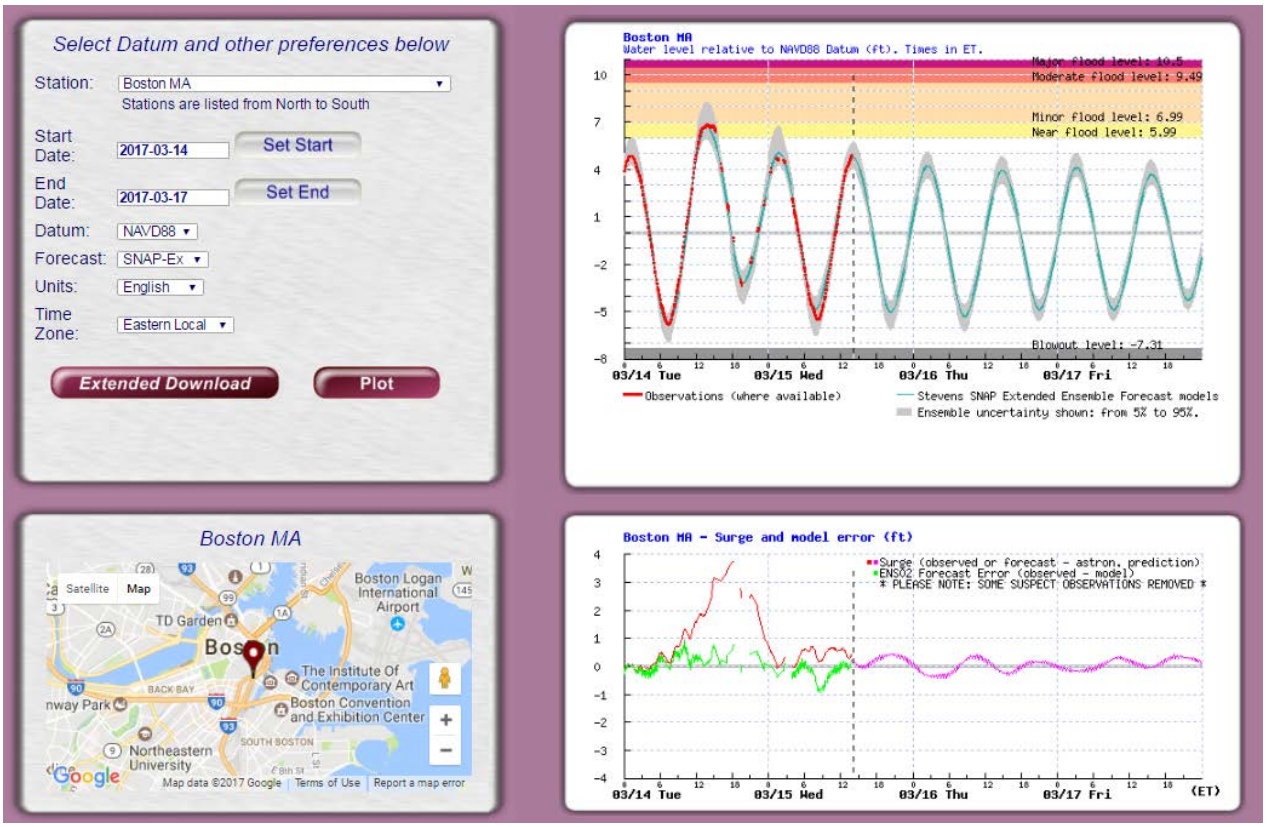


Newport RI



Percentiles of 96hr-prior forecast peaks.

Verified peak was 2.84ft NAVD88 on March 15 2017, 1:12 UTC.



Boston MA

Finally, Table 3 (next page) shows the observed or, in the absence of observations, simulated peaks for the storm.

Table 3. Preliminary Peak Water Levels for Nor'Easter Stella at Stevens Flood Advisory System Stations

Station Name	Peak Water Level (ft NAVD88)	Peak time (EDT)	NWS Flood Level
Duck NC	1.91	3/14/2017 21:06	
Chesapeake Bay Bridge Tunnel VA	1.55	3/14/2017 23:00	
Windmill Point VA	0.25	3/14/2017 11:24	
Lewisetta VA	0.73	3/14/2017 04:12	
Cambridge MD	0.26	3/14/2017 07:12	
Annapolis MD	-0.47	3/14/2017 06:48	
Baltimore MD	-0.88	3/14/2017 06:54	
Ocean City Inlet MD	1.76	3/14/2017 10:30	
Lewes DE	4.35	3/14/2017 10:30	Minor
Cape May NJ	4.59	3/14/2017 09:48	Minor
Great Channel at Stone Harbor NJ	4.27	3/14/2017 10:42	Minor
Maurice River at Bivalve NJ	3.98	3/14/2017 10:12	Minor
Cohancey River at Greenwich NJ	2.66	3/14/2017 00:42	
Ludlam Thorofare at Sea Isle City NJ	4.47	3/14/2017 10:36	Minor
Peck Bay at Ocean City NJ	2.77	3/14/2017 10:25	
Inside Thorofare Rte 40 Atlantic City NJ	4.61	3/14/2017 10:48	Minor
Atlantic City NJ	5.18	3/14/2017 09:18	Moderate
Absecon Creek Rte 30 at Absecon NJ	4.48	3/14/2017 10:54	Moderate
Little Egg Inlet near Tuckerton NJ	3.90	3/14/2017 09:54	Moderate
East Thorofare at Ship Bottom NJ	2.95	3/14/2017 12:48	Minor
Barnegat Bay at Barnegat Light NJ	3.06	3/14/2017 10:00	Minor
Barnegat Bay at Mantoloking NJ	0.96	3/15/2017 01:42	Near
Watson Creek at Manasquan NJ	4.39	3/14/2017 10:24	Moderate
Shark River at Belmar NJ	5.09	3/14/2017 10:18	Moderate
Shrewbury River at Sea Bright NJ	4.63	3/14/2017 12:06	Moderate
Sandy Hook NJ	5.06	3/14/2017 10:42	Moderate
Raritan Bay at Keansburg NJ	5.36	3/14/2017 11:24	Moderate
Raritan Bay at South Amboy NJ	5.04	3/14/2017 10:55	Minor
New York Harbor near Bay Ridge NY	4.73	3/14/2017 11:12	
New York Harbor near Bayonne NJ	4.69	3/14/2017 11:12	Near
New York Harbor near Red Hook NY	4.48	3/14/2017 11:30	Minor
The Battery NY	4.84	3/14/2017 11:18	Minor
Arthur Kill near Bloomfield NY	4.98	3/14/2017 11:36	Minor
Bergen Point West Reach NY	5.01	3/14/2017 11:36	Minor
Newark Bay near Elizabeth NJ	4.93	3/14/2017 11:48	
Elizabeth Channel near Newark NJ	5.10	3/14/2017 11:48	Minor
Passaic River at PVSC at Newark NJ	5.09	3/14/2017 12:00	Minor
Passaic River near Harrison NJ	5.02	3/14/2017 12:12	

Kearny Point at Kearny NJ	4.94	3/14/2017 12:00	Minor
Saw Mill Creek at Kearny NJ	5.00	3/14/2017 12:25	Minor
Barge Marina at Carlstadt NJ	6.02	3/14/2017 13:00	Moderate
Hackensack River at Hackensack NJ	5.00	3/14/2017 13:30	Minor
FDU Bridge at Hackensack NJ	6.02	3/14/2017 12:55	Minor
Hudson River at South Dock at West Point NY	3.27	3/14/2017 14:25	Near
Hudson River below Poughkeepsie NY	3.28	3/14/2017 15:15	Near
Norrie Point NY	2.51	3/14/2017 03:00	
Tivoli Bays South NY	3.13	3/14/2017 16:45	
Schodack Island NY	3.55	3/14/2017 06:00	
Hudson River at Albany NY	3.90	3/14/2017 06:15	
Hudson River at Piermont NY	3.70	3/14/2017 12:45	Near
Hudson Rv South of Hastings-on-Hudson NY	3.67	3/14/2017 12:35	Near
Hudson River at 39th St NY	3.98	3/14/2017 11:35	
Hudson River near Jersey City NJ	4.12	3/14/2017 11:25	Near
Hudson River near Hoboken NJ	4.53	3/14/2017 11:30	Minor
Hudson River near Hudson Square NY	4.52	3/14/2017 11:30	
Rockaway Inlet near Floyd Bennett Field NY	4.19	3/14/2017 10:45	
Bowery Bay at East Elmhurst NY	6.32	3/14/2017 12:36	Minor
Bergen Basin at Jamaica Bay NY	4.33	3/14/2017 11:36	Near
East Rockaway Inlet at Atlantic Beach NY	4.33	3/14/2017 10:36	Near
Jamaica Bay at Inwood NY	4.71	3/14/2017 11:42	Minor
Kings Point NY	6.76	3/14/2017 13:12	Moderate
Reynolds Channel at Point Lookout NY	4.21	3/14/2017 11:06	Minor
Hudson Bay at Freeport NY	4.03	3/14/2017 11:18	Moderate
Great South Bay at Lindenhurst NY	2.37	3/14/2017 12:30	Minor
Montauk NY	3.23	3/14/2017 13:48	Minor
Bridgeport CT	5.23	3/14/2017 15:30	Minor
New Haven CT	5.86	3/14/2017 14:18	Moderate
New London CT	3.21	3/14/2017 13:42	Minor
Newport RI	2.84	3/14/2017 21:12	
Woods Hole MA	2.42	3/14/2017 21:54	
Nantucket MA	2.85	3/14/2017 16:30	
Providence RI	3.50	3/14/2017 20:54	
Boston MA	6.85	3/14/2017 13:42	Near
Fort Point NH	5.84	3/14/2017 13:42	Near
Portland ME	5.83	3/14/2017 13:30	Near